

GENERAL PHARMACOLOGY

MBS230-P&T

UNIT 1.1.8

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1.1.8 ROUTES OF DRUG ADMINISTRATION

ROUTE OF DRUG ADMINISTRATION

LS 1.1.8: LEARNING OBJECTIVES:

Students should be able to:

- Describe the routes of drug administration, and stating their advantages and disadvantages

ROUTES OF DRUG ADMINISTRATION

INTRODUCTION:

The place in the body where absorption occurs will affect the process of absorption

The major routes of drug administration are:

- Enteral
- Parenteral
- Topical

ROUTES OF ADMINISTRATION

A. Enteral route

- **Oral ingestion**
- **Sublingual /buccal**
- **Intrarectal**

B. Parenteral route

- **Intravenous**
- **Intramuscular**
- **Subcutaneous**
- **intradermal**

ROUTES OF ADMINISTRATION

C. Other routes

- **Inhalational**
- **Intravaginal**
- **Topical**
- **Intranasal**
- **Intraocular**
- **Aural**
- **Transcutaneous**

DRUG ADMINISTRATION: ENTERAL ROUTES

- **Buccal and sublingual:** The drug is absorbed through the mucous membranes in the mouth
- **Oral:** The drug is swallowed and absorbed in the stomach and/or small intestine
- **Rectal:** The drug is introduced in the rectum and absorbed through the rectal mucosa

BUCCAL AND SUBLINGUAL ROUTES OF ADMINISTRATION

Mucous membranes of the mouth have a highly vascular absorbing surface and allow for absorption of drugs that are administered sublingual or buccal

Advantages: Quick effect is obtained, first pass metabolism is avoided and ability to swallow is not required.

Disadvantages: Absorptive area is small therefore only small amounts of drugs can be given, and inconvenient if use is frequent and if irritant to mucous membranes

ORAL ROUTE OF ADMINISTRATION

Absorption in the stomach

- Lipid soluble substances and those that are relatively non-ionized are well absorbed here
- The low pH (about 1.4), rate of gastric emptying and pH changes will affect how fast or how slow drugs are absorbed
- The stomach is not a significant absorption site unless the drug has characteristics that allow for rapid absorption
- Drugs absorbed in the stomach are susceptible to first pass metabolism

ORAL ROUTE OF ADMINISTRATION ... CONT'D

Absorption in the small intestine

- The small intestine is the most important site for absorption of oral drugs as it has extensive absorptive surface due to many villi, peristalsis and mixing encourage dissolution of drugs, it is highly vascular and has a pH of 7.0 to 8.0
- First-pass elimination may reduce bioavailability

ORAL ROUTE OF ADMINISTRATION ... CONT'D

Advantages:

- Economical
- convenient and acceptable to most patients.
- For drugs intended to have effect in the gut, the drug is placed at the site of action (e.g. neomycin, anti-helminths).

Disadvantages:

- The rate and extent of absorption is affected by a variety of factors

RECTAL ROUTE OF ADMINISTRATION

The aim of rectal administration is twofold:

- To produce local effect e.g. in ulcerative colitis, haemorrhoids
- To produce systemic effect e.g. diazepam, aminophylline

Avoids most first pass effects in the liver

The rectum has extensive vascularity but a limited surface area

RECTAL ROUTE OF ADMINISTRATION ... CONT'D

Advantages:

- Drugs that are irritant to the stomach can be given by suppository (e.g. aminophylline, indomethacin).
- Useful in patients who are vomiting or unable to take orally (e.g. postoperatively, uncooperative; unconscious).

Disadvantages:

- (1) Absorption is erratic and unpredictable
- (2) Rectal inflammation may occur with repeated use

PARENTERAL ROUTES OF ADMINISTRATION

Include:

- Intravenous (IV)
- Intramuscular (IM)
- Subcutaneous (SC)
- Inhalation
- Intradermal
- Transdermal
- Intra-articular (synovial cavity)
- Intrathecal
- Intraperitoneal

IV ROUTE OF ADMINISTRATION

Given as bolus or IV infusion. The drug must be in soluble form and there is need for asepsis.

Advantages

- The fastest and most certain route of administration with 100% bioavailability
- Suitable for administration of drugs that are not absorbed from the gut or are too irritant to be given by other routes (e.g. anti-cancer drugs)
- The magnitude and duration of response of a drug can be controlled accurately when given by IV infusion thus making this route particularly useful in the treatment of emergencies when absolute control of drug administration is essential

IV ROUTE OF ADMINISTRATION CONT'D

Disadvantages

- Hazardous if given too quickly
- Local venous thrombosis with prolonged infusion and with bolus doses of irritant drugs
- Infection of the intravenous catheter may occur with prolonged infusion

IM AND SC ROUTES OF ADMINISTRATION

- After IM and SC administration, many drugs can enter the capillaries directly through pores between endothelial cells
- Usually produce a faster effect than oral administration
- Absorption is affected by diffusion through the tissue and local blood flow

IM AND SC ROUTES CONT'D

- Intramuscular route is more effective than the subcutaneous route because there is a greater blood supply in muscle tissue.
- Application of heat or massage can increase vasodilatation and improve absorption

IM AND SC ROUTES CONT'D

Advantages:

- (1) IM is reliable and suitable for some irritant drugs
- (2) Depot preparations can be used with IM administration
- (3) SC is reliable and acceptable for self-administration

Disadvantages:

- (1) IM is not acceptable for self-administration, may be painful, and if any adverse effects occur to a depot formulation, it cannot be removed
- (2) With SC there is poor absorption in peripheral circulatory failure, and repeated injections at one site can cause lipodystrophy

INTRATHECAL ROUTE OF ADMINISTRATION

This is injection of a drug into the subarachnoid space or epidural space via a lumbar puncture needle e.g. methotrexate (to prevent relapse of leukaemia in the CNS), bupivacaine (a local anaesthetic), opiate analgesics, baclofen, and antibiotics such as aminoglycosides which do not cross the blood brain barrier

INHALATION (PULMONARY) ROUTE OF ADMINISTRATION

- General anaesthetic gases and vapours, and aerosols can be delivered by this route
- The aim is either to provide a local effect (e.g. bronchodilatation) or systemic effects (general anaesthesia)
- Rapid absorption occurs due to large surface area, rich blood supply and high permeability of the alveolar membrane

INHALATIONAL ROUTE OF ADMINISTRATION ... CONT'D

Volatile and gaseous anaesthetics when given by inhalation can be taken up and eliminated rapidly thus giving close control in general anaesthesia

In the treatment of bronchial asthma, this route allows for higher drug concentrations to be achieved in the lungs than in systemic circulation, thus minimizing systemic side effects. Self-administration is practicable with aerosols.

Disadvantages

- Special apparatus is required
- The drug must be non-irritant if the patient conscious

TRANSDERMAL ROUTE OF ADMINISTRATION

A disk or patch containing a days' or weeks' medication designed to be absorbed at a steady rate

Examples: Oestrogen for hormone replacement, glyceryl trinitrate

Advantages: Produces a steady rate of drug delivery and avoids presystemic metabolism

Disadvantages: (1) Suitable only for relatively lipid-soluble drugs (2) Transdermal patches are expensive

TOPICAL ROUTES OF ADMINISTRATION

Used when a local effect is desired (e.g. in the oral cavity, skin, eye, lung, vagina and rectum) or to minimize systemic effects

Advantage: High local concentrations are achieved without systemic effect

Disadvantage: Absorption may occur giving systemic effects

LEARNING OUTCOMES

Students must be able to comprehend:

The routes of drug administration, and the advantages and disadvantages of each the most commonly used routes of administration